LOAD LIFTING

Manual Load Lifting

Manual handling is any activity requiring the use of force, exerted by a person to lift, lower, push, pull or carry or otherwise move, hold or restrain an object, person or animal. Lifting and moving loads manually is one of the most common causes of injury at work, 56,000 Work Related MSD cases in construction industry per year. There is no truly 'safe' weight limit for manual handling operations so try to find alternatives. Manual handling includes lifting, putting down, pushing, pulling, carrying, moving or supporting a load by hand or bodily force.

Hazardous manual handling refers to activities that have:

- Repetitive or sustained application of force
- Repetitive or sustained awkward posture
- Repetitive or sustained movement
- Application of high force
- Exposure to sustained vibration
- Handling live persons or animals.

What are the risks in manual load lifting?

Work related MSD may include injuries to:

- Muscles
- Ligaments
- Intervertebral disc
- Nerves
- Tendons in the wrist, arms, shoulders, neck or legs

Causes of injury in manual load lifting

- The weight of the load
- The size and shape
- The available grip
- The way that the load is carried, where and how often it has to be carried, and over what distance

Workers may suffer from musculoskeletal problems such as aches, strains and sprains as a result of manual handling. These can also be caused by other tasks which involve repetitive movements, force, unusual postures, and prolonged pressure on a joint, badly organized working practices or work environment.

Effects on health can include: sprains or strains backache sciatica hernias arthritis swelling of the wrist, forearm, elbow and shoulder

Avoid manual handling by;

•**Better work planning** can remove the need for loads to be moved more than once. For example, deliver a bagged load directly on to a trailer to remove the need for handling

•carrying out tasks near the load rather than moving the load to the task

•full mechanization of the task

Assess the risks from loads that:

•weigh over 25kg or are difficult to grip or handle

•need frequent lifting or lowering are difficult to maneuver,

•and/or involve twisting or lifting above shoulder height or from the floor

Reduce the risk of injury by:

•using mechanical assistance, for example, sack trucks

•changing the type of load, for example, 20kg bags rather than 50kg

•finding improved ways of handling, for example, reducing the risks from animal handling by properly designed cattle races and crushes

•providing training for workers where manual handling cannot be avoided

Why is dealing with manual handling important?

Manual handling injuries can have serious implications for the employer and the person who has been injured. They can occur almost anywhere in the workplace and heavy manual labour, awkward postures, repetitive movements of arms, legs and back or previous/existing injury can increase the risk.

Importance of improving your workplace during load lifting

- ✓ Reducing or preventing injuries
- ✓ Reducing workers efforts
- ✓ Reducing MSD risk factors
- ✓ Increases productivity
- ✓ Lowering costs related to medical services of injured staffs by reducing production bottle necks, errors rates or rejects, workers' compensation claims.

For any lifting activity always take into account:

•individual capability

•the nature of the load

•environmental conditions

•training

•work organization

If you need to lift something manually

•Reduce the amount of twisting, stooping and reaching

•Avoid lifting from floor level or above shoulder height, especially heavy loads

•Adjust storage areas to minimize the need to carry out such movements

•Consider how you can minimize carrying distances

•Assess the weight to be carried and whether the worker can move the load safely or needs any help – maybe the load can be broken down to smaller, lighter components

If you need to use lifting equipment

•Consider whether you can use a lifting aid, such as a forklift truck, electric or hand-powered hoist, or a conveyor

•Think about storage as part of the delivery process – maybe heavy items could be delivered directly, or closer, to the storage area

•Reduce carrying distances where possible

Good handling technique for lifting

There are some simple things to do before and during the lift/carry:

•Remove obstructions from the route.

•For a long lift, plan to rest the load midway on a table or bench to change grip.

•Keep the load close to the waist. The load should be kept close to the body for as long as possible while lifting.

•Keep the heaviest side of the load next to the body.

•Adopt a stable position and make sure your feet are apart, with one leg slightly forward to maintain balance

Guide line for safe lifting

- Start in a good posture. At the start of the lift, slight bending of the back, hips and knees is preferable to fully flexing the back (stooping) or fully flexing the hips and knees (squatting).
- Don't flex the back any further while lifting. This can happen if the legs begin to straighten before starting to raise the load.
- Keep the load close to the waist. Keep the load close to the body for as long as possible while lifting. Keep the heaviest side of the load next to the body. If a close approach to the load is not possible, try to slide it towards the body before attempting to lift it.
- Avoid twisting the back or leaning sideways, especially while the back is bent. Shoulders should be kept level and facing in the same direction as the hips. Turning by moving the feet is better than twisting and lifting at the same time.
- Keep the head up when handling. Look ahead, not down at the load, once it has been held securely.
- Move smoothly. The load should not be jerked or snatched as this can make it harder to keep control and can increase the risk of injury.
- Don't lift or handle more than can be easily managed. There is a difference between what people can lift and what they can safely lift. If in doubt, seek advice or get help.
- Put down, then adjust. If precise positioning of the load is necessary, put it down first, then slide it into the desired position.

What do you have to do?

1. **IDENTIFY MANUAL HANDLING TASKS**
   - Talk to colleagues
   - Conduct a walk through the workplace
2. **ASSESS THE LEVEL RISK ASSOCIATED WITH THE TASK**
   - Document manual handling in the OHS Risk Register
   - Complete risk assessment where the tasks or level of risk is uncertain
3. **IDENTIFY AND IMPLEMENT CONTROLS**
   - Talk to colleagues undertaking task
   - Implement risk controls
   - Document risk controls
   - Report all manual handling incidents on eduSafe

Examples of risk control

- ✓ Don't undertake the task
- ✓ Change the environment
- ✓ Change the nature of the work
- ✓ Change the object
- ✓ Use mechanical aids
- ✓ Administrative aids (coaching, mentoring and goals)

S.M.A.R.T Lifting

Team Lifting

Consider:

- Are there enough people?
- Are all persons of the same size with similar strength?
- Are there any known pre-existing injuries?
- Who is coordinating the lift?
- Is there a plan, and has been communicated to those involved?

Pushing and Pulling

- When pushing, lean forward
- Stand close to the trolley
- When pulling, lean backwards (**pushing is preferable** as it involves less work by the lower back muscles and allows for maximum use of body weight).
- providing clearer vision in the direction of travel
- Ensuring you have a good grip
- Avoiding twisting and turning
- Checking that the handle height is between shoulder and waist height

Safe Storage

Consider:

- Storing frequently and heavy items between waist and shoulder height
- Storing smaller, lightweight or infrequently used items in lower or higher areas
- Knowing shelving weight limits
- Having access to a step-ladder
- Testing the weight of the object before picking it up
- Conducting quarterly workplace inspections

POSTURE & ANGLES

Posture is the position and carriage of the body as a whole. Let's look at some of the correct and incorrect positions and angles. These are positions that the employees need to be informed of or trained on. They do not necessarily come to the employees naturally.

Good posture means that the key parts of a person's body are correctly aligned and supported by the right amount of muscle tension.

BODY ANGLES;

Head- Should be straight and balanced over the spine while looking forward. Eliminate the flexed-neck position. Center the monitor in front of you and position the top of the monitor 2" to 3" above seated eye level. Keep your neck in a neutral or aligned position.

Shoulders- Keep level and straight and relaxed. If the chair has armrests make sure are adjusted correctly so are not hunching up your shoulders. Adjust chair so back is supported with proper lumbar curve.

Elbows- Keep elbows is a slightly open angle (100-110 degrees) with your wrists is a straight position. A keyboard that tilts can help you keep the correct position if changing the position of the chair. Keep the mouse within close reach so do not have to stretch to use it. Adjust the height of the armrests so they allow the user to rest arms at their sides and relax their shoulders while keyboarding. Elbows and lower arms should rest lightly so as not to cause circulatory or nerve problems.

Hands and wrists- Your hands should be slightly lower than your elbows. Keep wrists neutral. The safe zone of movement for your wrist is about 15 degrees in all directions.

Waist- Straight and not twisted. Keep at 90-120 degree angle. Hips should be against the back of the chair.

Legs- Thighs should be parallel to the floor and the knees at about the same level as the hips. If you feel pressure on the back of the thigh, lower the chair slightly. Back of knees 3" to 4" away from chair to prevent the seat from cutting off circulation to the lower legs. You should be able to easily slide your fingers under the front area of the thigh. The chair should be long enough and wide enough to support your hips and thighs. Not all ergonomic chairs are made the same.

Feet- Place your feet flat on the floor. Use a foot rest if needed

Correct posture can help by:

- reducing strain on the body during movement and exercise
- reducing wear and tear on the joints, muscles, and ligaments
- maintaining balance while moving and exercising
- reducing the risk of muscle strain and overuse conditions
- improving spine health

Best sitting position

The best sitting position depends on a person's height, the chair they are using, and the activity they are doing while sitting. A person can improve posture and achieve a proper sitting position by:

- keeping feet flat or rest them on either the floor or a footrest
- avoiding crossing knees or ankles
- maintaining a small gap between the back of the knees and the chair
- positioning knees at the same height or slightly lower than the hips
- placing ankles in front of the knees
- relaxing the shoulders
- keeping the forearms and knees parallel to the floor where possible
- holding elbows at the sides creating an L-shape in the arms
- sitting up straight and looking forward without straining the neck
- keeping the back against the chair, or using a backrest or cushion if there are places where the back does not comfortably meet the chair, especially in the lower back region
- avoiding sitting for long periods at a time, ideally taking at least a 10-minute break for every hour of sitting

Sitting positions to avoid

Anything that causes a misuse or overuse of specific muscles, ligaments, or tendons can adversely impact a person's posture and back health. Certain positions are worse than others for overworking or misusing postural tissues, especially some sitting positions.

To prevent bad posture and back health avoid:

- sitting slumped to one side with the spine bent
- keeping the knees, ankles, or arms crossed
- dangling or not properly supporting the feet
- sitting for a long time in one position
- straining the neck for long periods while looking at a monitor, telephone screen, or document
- sitting in a position that does not fully support the back, especially the lower back
- sitting for an extended period without taking a break

When standing:

- Bear your weight primarily on the balls of your feet.
- Keep your knees slightly bent.

- Keep your feet about shoulder-width apart.
- Let your arms hang naturally down the sides of the body.
- Stand straight and tall with your shoulders pulled down and backward.
- Tuck your stomach in.
- Keep your head level. Your earlobes should be in line with your shoulders. Do not push your head forward, backward, or to the side.
- Shift your weight from your toes to your heels, or one foot to the other, if you must stand for a long time.

Steps to Good Walking Posture

- **Stand up straight**. Visualize being tall and straight, like a tree. Do not arch your back.
- **Do not lean forward or lean back**. Leaning puts a strain on the back muscles as you walk, and you should avoid leaning except when on a hill. Unfortunately, some coaches have taught leaning for speedwalking, but it is not necessary, and often people lean too far forward or back.
- **Keep your eyes forward**. Avoid looking down. Your focus should be about 20 feet ahead of you. This way you will be able to see your path and anything coming at you from the sides.
- **Keep your chin up (parallel to the ground).** This reduces strain on your neck and back. A proper chin position will also allow you to look ahead rather than down at your feet.
- **Let your shoulders be back and relaxed**. Shrug once and allow your shoulders to fall and relax, slightly back. Loosening up the shoulders in this way will help relieve tension and put them into a position to use good arm motion while walking. You can also do this at intervals during your walk to ensure you are keeping your shoulders relaxed.
- **Suck in your stomach**. Your core muscles can help you maintain good posture and resist slouching and leaning. Keeping your stomach pulled in slightly (while still taking deep, full breaths) can help you maintain good walking posture.2
- **Tuck in your behind and rotate your hips forward slightly**. This will keep you from arching your back. You shouldn't have your butt sticking out while you're walking.

The risk of injury can increase when working overhead for long periods of time, especially when holding tools, equipment or materials above shoulder height. Twisting your body while your arms are in an awkward position, or repetitive movements combined with the use of force can also add to the risk, as can lifting, holding and positioning heavy or awkward objects while your arms are raised.

Overhead work is a common part of our day-to-day work, and the good news is it can be performed safely. There are solutions available that can reduce the level of stress on your shoulders, neck and arms, as well as reduce how often and how long the body is subjected to this stress. Many of the solutions can also eliminate other potential safety hazards and increase productivity.

The type of task and the site conditions will determine which solutions are best for you. A few possible solutions for specific overhead tasks are:

• Change materials or work processes. Use materials, building components or work methods that are less labor-intensive so the task takes less time and you reach overhead for a shorter period.

• Change tools and/or equipment. For example, use bit extensions for drills and screw guns that allow you to hold the tool at waist or shoulder level rather than above your head. Use mechanical lifts or hoists to raise and position yourself closer to the work.

• Change work rules and provide training. Workers need to be made aware of the hazards and be trained on the safest way to perform their work. They should be encouraged to utilize every resource available to them to make overhead work a safe experience

Safe Stacking and storage

Improper stacking and storage can result in injuries to workers and damage to costly materials

Although OSHA does not provide much specific direction concerning safe stacking and storage, 29 CFR 1910.176(b) of the material handling standard does generally require secure workplace storage of materials. The regulation states: "Storage of material shall not create a hazard. Bags, containers, bundles, etc., stored in tiers shall be stacked, blocked, interlocked and limited in height so that they are stable and secure against sliding or collapse."

General safe stacking and storage rules with which employees who handle materials should be familiar include:

- Ensure that stacks are stable and self-supporting.
- Observe height limitations when stacking materials.
- Allow sufficient clearance around stacks for safe handling and easy access.
- Make sure there is sufficient clearance between stacks and lights, heating pipes, and sprinkler heads.
- Make sure stacks don't block emergency exits, emergency equipment, or fire alarms.

When stacking boxes, employees should:

- Place boxes on a pallet for stability and to make them easy to move.
- Interlock boxes to make a more stable stack.
- Band boxed materials or secure them with cross-ties or shrink wrap.

- A good rule of thumb for ensuring a stable stack is to observe a height to base ratio that does not exceed 3:1 (or 4:1 at most if the stack is effectively interlocked and there is a good grip between the contacting surfaces).

When stacking bags, sacks, and baled and bundled materials, workers should:

- Stack bags and bundles in interlocking rows to keep them secure.
- Stack bagged material by stepping back the layers and cross-keying the bags at least every 10 layers.
- Store baled paper and rags inside a building no closer than 18 inches to the walls, partitions, or sprinkler heads.

When stacking pipes, poles, and bars, employees should:

- Not store pipes, poles, and bars in racks that face main aisles to avoid creating a hazard to passersby when removing supplies.
- Stack and block pipes and poles as well as structural steel, bar stock, and other cylindrical materials to prevent spreading or tilting unless they are in racks.

When stacking barrels and drums, workers should:

- Stack drums, barrels, and kegs symmetrically.
- Chock the bottom tiers of drums, barrels, and kegs to keep them from rolling if stored on their sides.
- Place planks, sheets of plywood dunnage, or pallets between each tier of drums, barrels, and kegs to make a firm, flat stacking surface when stacking on end.

When stacking lumber, employees should:

- Stack lumber no more than 16 feet high if it is handled manually, and no more than 20 feet if using a forklift.
- Remove all nails from used lumber before stacking.
- Stack and level lumber on solidly supported bracing.

Stable construction of the stack is entirely dependent upon the following factors:

— Safe relation of height to dimension of base;

— Sound interlocking of the goods; — Contents of the sacks or cartons;

— Compact construction and avoidance of transfer stress in any dunnage used for artificial bonding;

— Shape of articles e.g. meat carcasses;

— Determination of the aggregate weight to be borne by the components in the lowest tier of the stack; — Good placing of every component forming part of the stack, with special care taken to avoid overhang on any side or end of a stack intended to be vertical..

Basic rules for breaking down stacks (De-stacking) are:

- One person only should be responsible for the manner in which the stack is reduced. This is particularly important where gangs are employed.
- If the person in charge of destacking had no part in erecting the stack, he or she should ascertain its construction before work begins.
- The stack should be taken down tier by tier, so that no part is in danger, as would happen if the stack were bitten into locally.
- When a large stack has been given a natural fence by building the periphery higher than the centre, this fence should be maintained during destacking. The work should be done outward from the centre, reducing the periphery last so that there is a barricade to prevent people falling off. – If tubular or other fencing has been built in or around the stack, it should be dismantled and adjusted as the height is decreased.
- As there is a high tripping hazard in the working area of a stack, tidiness and systematic work methods are essential. Care should be taken to ensure that people working on stacks are at all times clear of overhead travelling cranes. Proper signaling arrangements should be made and a lookout should be posted. All stacking and destacking should be carried out under competent supervision.

SIMPLE EXERCISES TO RELIEVE WORK STRAIN

Neck Stretch; Drop your head to one side and hold for 10 seconds. Repeat to the other side. Drop your head to your chest and hold 10 seconds. Turn head slowly from side to side.

Chin tucks; Raise the head to straighten the neck. Tuck the chin in and upwards creating a double chin. This also results in a forward tilt of the head. Repeat several times.

Upper and lower back stretch; Interlace fingers and turn palms upwards above head; straighten arms then slowly lean slightly from side to side. Repeat movement several times.

Back arching; Stand up. Support your lower back with hands and gently arch back and hold for 5 to 10 seconds. Repeat as often as is needed.

Pectoral stretch; Raise both arms to shoulder height and bend elbows. Pull both elbows back slowly to bring shoulder blades towards each other.

Shoulder Shrug and Roll; raise your shoulders toward your ears. Hold and release. Roll shoulders forward and repeat in the opposite direction.

Finger Fan; Spread you fingers wide with palms down and hold 6 seconds. Make a tight fist and release.

Wrist and elbow stretch; Interlace fingers, palms outward, and straighten arms in front. Hold for 10 seconds and repeat several times.

Wrist stretch; Straighten your arm in front and bend your wrist forward, gently assist the stretch with your other hand.

Hold for 10 seconds then stretch your wrist back and hold for 10 seconds. Repeat with other arm.